Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Decentralized Swarm Intelligence for Resilient Manufacturing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity (), AI (✓), Next-gen Communication ( ), Advanced Robotics & Manufacturing (✓), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Robotics & Manufacturing		Artificial Intelligence	
	Medium Category	Small Category	Medium Category	Small Category
	Advanced autonomous control	Autonomous robots	Industrial use and innovative AI	Advanced AI modeling / decision-making (cognition / judgment / inference)
Concept of Proposed Technology	Small and medium-sized manufacturing enterprises increasingly rely on custom part manufacturing and small-batch, high-variation production to remain competitive. However, existing solutions impose significant barriers: independent producers face high upfront automation costs, while centralized manufacturing execution systems (MES) are expensive, rigid, and poorly suited to fluctuating demand, heterogeneous equipment, and rapid reconfiguration. These limitations make it difficult for SMEs to scale production, collaborate across sites, or respond resiliently to disruptions. This project addresses these challenges by developing Swarm Contracts, a decentralized coordination framework co-developed by the project proposer in prior work (Grey et al., 2020; Grey et al., 2021; Mallikarachchi et al., 2022). Swarm Contracts enable manufacturing tasks to be			

	dynamically decomposed, allocated, executed, verified, and settled by autonomous agents, without relying on a central controller. At a high level, Swarm Contracts operate as follows: manufacturing clients post-production tasks (e.g., machining, assembly, inspection); autonomous robotic and cyber-physical workers, such as autonomous mobile robots (AMRs), robotic arms, or specialized processing units, bid on sub-tasks based on their capabilities, availability, and local operational context. Once a task is completed, independent adjudicators, implemented through algorithmic verification or sensing-based validation modules, verify compliance with specifications before triggering payment and reputation updates. Unlike traditional centralized MES platforms, Swarm Contracts enable plug-and-play participation by heterogeneous equipment and organizations. SMEs can incrementally adopt the system, adding new machines, subcontractors, or verification modules, without redesigning their entire production pipeline or trusting a single controlling entity. The resulting decentralized, incentive-driven manufacturing architecture offers several advantages that are particularly valuable to SMEs: • Adaptive scheduling that responds to machine availability, failures, and demand changes in real time • Trustworthy verification of work quality without requiring prior trust between parties • Fair, transparent incentives that ensure participants are paid for verified contributions • Operational resilience, allowing production to continue despite partial system failures • Low-cost experimentation, enabling SMEs to iteratively test automation and collaboration strategies before full-scale deployment Through this project, we aim to advance Swarm Contracts from a validated research concept to a practical coordination layer for resilient, decentralized manufacturing, addressing the following core challenges: • **Decentralized Scheduling:** How can manufacturing agents coordinate tasks efficiently without a central manager? • **Verification of Work:** How can completed parts be validated against specifications in a trustworthy and automated manner? • **Incentive Alignment:** How can fair compensation and accountability be ensured across independent participants? • **Scalability:** How can the system grow in number of agents and tasks while remaining efficient? • **Provability and Testability:** How can SMEs evaluate the system incrementally, at low cost and low risk?
Rationale for Joint R&D Project	Manufacturing environments, particularly those supporting small-batch, high-variety production, are becoming increasingly dynamic, distributed, and heterogeneous. While centralized manufacturing execution and scheduling systems remain dominant, they exhibit fundamental limitations in such settings, including: • Limited adaptability to unpredictable demand, machine failures, and supply-chain disruptions

- High integration and maintenance costs, particularly when incorporating heterogeneous robotic systems
- Single points of failure that reduce operational resilience
- Difficulty coordinating production across independently owned or geographically distributed assets and sites

These challenges are especially acute for small and medium-sized manufacturing enterprises. Unlike large manufacturers, these organizations often lack the capital and organizational flexibility required to deploy and maintain centralized automation infrastructure. As a result, SMEs face:

- High upfront investment barriers to automation
- Limited ability to experiment with flexible, decentralized production models
- Persistent integration challenges when deploying robots and equipment from multiple vendors

Addressing these constraints requires a fundamentally different coordination paradigm, one that supports decentralization, heterogeneity, and incremental adoption. The proposed Joint R&D project responds to this need by advancing Swarm Contracts, a decentralized coordination framework developed by the U.S. research team at RPI, into industrial manufacturing environments through close collaboration with Korean SMEs.

The project proposer at RPI has established the foundational technical framework for Swarm Contracts through extensive simulation studies and physical robotic deployments, demonstrating decentralized task allocation, incentive mechanisms, and verification protocols under heterogeneous agent behavior. However, transitioning this technology from laboratory-scale prototypes to real-world manufacturing systems requires deep integration with industrial-grade robotic platforms, factory workflows and operational constraints, capabilities that reside with Korean manufacturing SMEs.

Conversely, Korean SMEs are actively seeking flexible, resilient automation solutions that reduce dependency on rigid centralized systems and enable collaboration across organizational boundaries. Through joint R&D, Korean partners will directly influence the design of decentralized coordination protocols, ensuring alignment with practical requirements such as safety constraints, production tolerances, latency, reliability, and cost efficiency. Therefore, this joint project is essential to:

- Translate Swarm Contracts from research prototypes to deployable industrial systems
- Co-design decentralized coordination mechanisms grounded in real manufacturing workflows
- Enable Korean SMEs to adopt and shape next-generation manufacturing technologies

Key Words	1. Decentralized Coordination 2. Distributed Autonomous Systems 3. Smart Manufacturing 4. Collaborative Robots (Cobots) 5. Flexible Manufacturing Systems

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2)	**End of the Project**	Level (6)
R&D Objectives	1. **Develop an industrial-grade Swarm Contract framework** that enables decentralized task decomposition, bidding, execution, verification, and settlement among heterogeneous robotic and cyber-physical agents in manufacturing environments. 2. **Enable decentralized scheduling and coordination under uncertainty**, allowing autonomous manufacturing agents to adapt in real time to dynamic demand, machine availability, and disruptions without reliance on centralized control systems. 3. **Implement automated verification and quality assurance mechanisms** that validate completed manufacturing tasks against specifications before settlement, ensuring trustworthy collaboration across independent participants. 4. **Design incentive and reputation** models that align agent behavior with manufacturing objectives, ensure fair compensation for verified work, and support sustainable participation by SMEs and independently owned robotic assets. 5. **Validate the Swarm Contract system** in realistic SME manufacturing scenarios, integrating industrial robotic platforms and evaluating performance, scalability, resilience, and cost efficiency against centralized baselines. 6. **Establish deployment and commercialization pathways** for SMEs, including reference architectures, deployment guidelines, and technology transfer strategies to support real-world adoption beyond the project period.			
R&D Target Specifications	1. **Scalability:** Support coordination of 50–500 heterogeneous robotic agents and maintain stable performance as agents dynamically join and leave the system. 2. **Reliability & Fault Tolerance:** Maintain ≥ 95% task completion rate under normal conditions and continue operation with up to 49% agent or network failure, enabling graceful degradation without system-wide shutdown. 3. **Decentralization & Autonomy:** Eliminate single points of failure in coordination logic and enable agents to make decisions using local state and shared protocols only. 4. **Interoperability:** Support robots from multiple vendors and control platforms and integrate with existing factory systems, while supporting standard industrial communication protocols (e.g., OPC-UA, MQTT, REST). 5. **Adaptability & Robustness:** Re-plan tasks within ≤ 30 seconds after disruptions or failures, while operating effectively in high-mix, low-volume production environments and handle partial observability and noisy sensor data. 6. **Security & Trust:** Authenticate and authorize participating robotic agents to prevent unauthorized modification of coordination protocols and maintain auditable logs of coordination and task execution. Also minimize data sharing to protect proprietary operational information. 7. **Deployment & Validation:** Validate system performance in large-scale simulations across several scenarios to demonstrate functionality on physical industrial robotic platforms and compare performance against centralized and heuristic baselines.			

R&D Plan

Phase I (6 months): Foundational Protocol Design and Validation

In phase 1, we will carry out the foundational protocol design, simulation-based validation, and quantitative performance evaluation.

1. Swarm Contract Protocol Design

We will formalize and extend the Swarm Contract protocol that enables decentralized task coordination without a central manager. Building directly on the proposer's prior Swarm Contract research, this phase focuses on protocol rigor, correctness, and the application to a smart manufacturing environment. Key activities include:

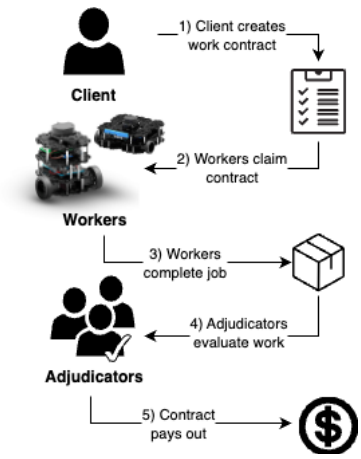

Fig 1: Swarm Contract Protocol

1. Formal specification of the Swarm Contract lifecycle: task posting → bidding/claiming → execution → adjudication → settlement, as seen in fig 1.
2. Smart contract mechanisms for incentive alignment among autonomous agents, and trust-minimized dispute resolution and adjudication
3. Explicit definition of worker roles, including producers, transporters, processors, verifiers.

This work establishes a clear and auditable protocol foundation suitable for later physical and industrial deployment.

2. Simulation-Based Manufacturing Environment

To enable low-cost, repeatable experimentation, we will extend an existing Swarm Contracts simulator to model realistic manufacturing workflows. Simulation enhancements include explicit representation of:

- Materials, parts, and intermediate production artifacts
- Multi-step production processes
- Modeling of heterogeneous worker capabilities, such as: maximum load capacities, movement speeds, processing abilities and constraints

We will ensure the following aspects within the simulation:

- Autonomous robots manufacture custom parts from colored material as can be seen in fig 2.
- Multiple processing paths are available to convert materials into parts
- Workers specialize in transportation, processing, or hybrid roles
- Complex parts require collaborative execution among multiple workers

This environment enables controlled evaluation of decentralized coordination strategies prior to physical deployment.

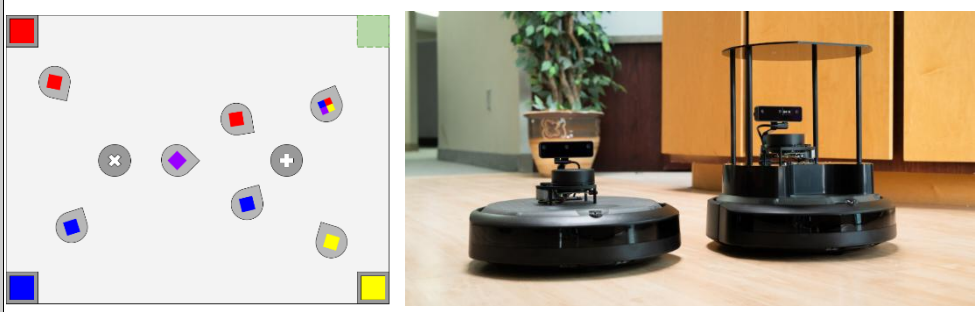

Fig 2: Simulation and Protocol Testing in Laboratory Setting

3. Performance Metrics and Baseline Evaluation

Evaluation metrics include:

- Reduction in worker idle time
- Task completion latency
- Resource utilization efficiency
- System robustness under worker heterogeneity

As can be seen in Fig 3, preliminary results already indicate that Swarm Contracts significantly reduce idle time among workers. Also, our prior work indicates that the protocol is robust to heterogeneous agent capabilities (speed, load, processing ability), and even adversarial conditions (Grey et al. 2020). We will further demonstrate productive and trustworthy collaboration among autonomous agents in protocol extensions.

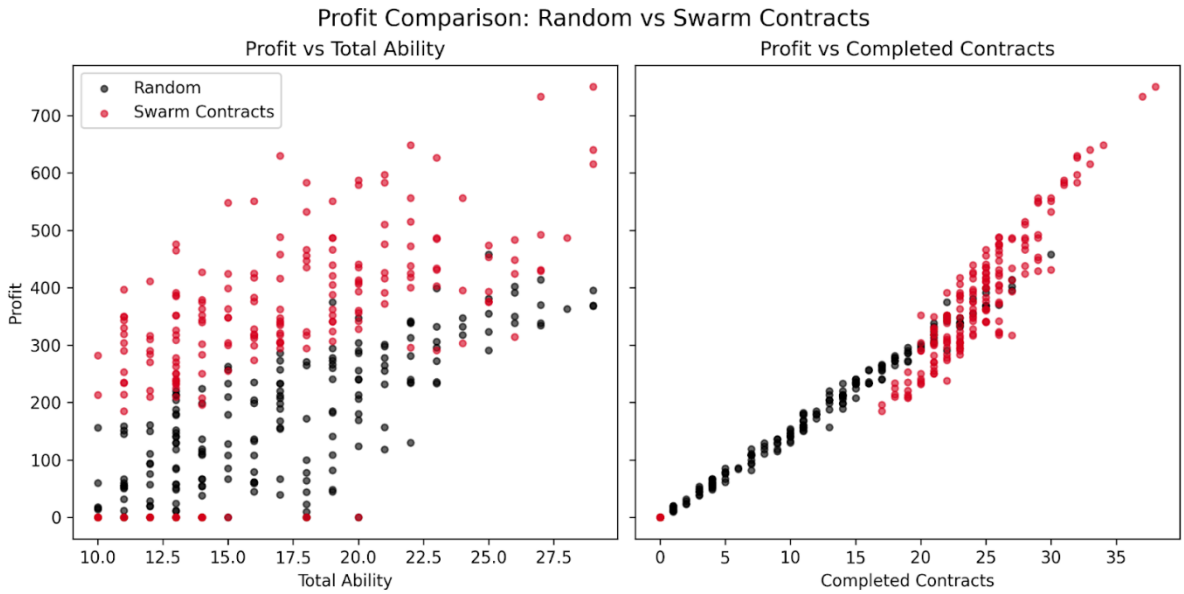

Fig 3: Preliminary Results

Phase II (2.5 Years): Physical Deployment and Industrial Integration

This phase positions Swarm Contracts for adoption as a next-generation decentralized coordination framework for smart manufacturing. Essentially, phase II extends phase I's the validated protocol into physical robotic systems.

4. Physical Robotic Testbed Integration

This task transitions Swarm Contracts from controlled simulations to real-world embodied systems.

Key activities include the deployment of Swarm Contracts on physical robotic platforms such as:

- Autonomous Mobile Robots (AMRs)
- Collaborative robots (cobots)
- Integration of sensing and verification mechanisms to support task completion validation, automated adjudication, and the demonstration of multi-robot collaboration on complex, multi-step manufacturing tasks.

5. Advanced Coordination and Scalability

To support realistic manufacturing environments, Phase II introduces advanced coordination mechanisms, including:

	• Contract splitting and merging for complex tasks
	• Dynamic adaptation to changing task demands, robot availability and failures, and scalability evaluation with increasing numbers of agents and concurrent contracts
	• These enhancements ensure that the system remains efficient, robust, and flexible under industrial-scale workloads.
	
	6. Industrial Pilot Deployment and Validation
	
	The final stage of Phase II focuses on real-world validation with a Korean SME manufacturing partner, where the activities include:
	
	• Pilot deployment in an operational manufacturing environment
	• Evaluation of system robustness under real-world disturbances, such as:
	  ○ Equipment downtime
	  ○ Communication delays
	  ○ Demand variability
	• Collection of operational feedback to inform:
	  ○ Protocol refinement
	  ○ Integration guidelines
	  ○ Commercialization readiness

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	RPI will serve as the scientific and technical lead, responsible for the design, validation, and advancement of the core Swarm Contract technologies.
	Key responsibilities include:
	• **Core Protocol Research and Design**
	  ○ Formal specification of the Swarm Contract lifecycle and coordination logic
	  ○ Design and verification of smart contract mechanisms for incentive alignment, adjudication, and settlement
	  ○ Development of advanced coordination features (contract splitting/merging, dynamic adaptation)
	• **Algorithm Development and Validation**
	  ○ Design of decentralized coordination algorithms for heterogeneous autonomous agents
	  ○ Analysis of robustness, scalability, and fault tolerance under decentralized control
	  ○ Definition of performance metrics and evaluation methodologies
	• **Simulation Environment Development**
	  ○ Extension and maintenance of the Swarm Contracts simulation framework
	  ○ Modeling of realistic manufacturing workflows, agent heterogeneity, and disruption scenarios
	  ○ Benchmarking against centralized and heuristic baselines
	• **Experimental Evaluation and Analysis**
	  ○ Design and execution of large-scale simulation experiments
	  ○ Quantitative analysis of system performance, efficiency, and resilience
	  ○ Documentation of experimental results and lessons learned
	• **Physical System Prototyping Support**
	  ○ Development of reference implementations for physical robotic platforms
	  ○ Integration support for sensing, verification, and adjudication mechanisms
	  ○ Collaboration with the SME partner on protocol deployment and debugging
	• **Knowledge Dissemination and IP Contribution**
	  ○ Publication of research results in peer-reviewed venues
	  ○ Contribution to joint intellectual property, patents, and technical documentation
	  ○ Training and technical guidance for SME engineers on decentralized coordination concepts

Expected R&R of Korean SME Partner	The Korean SME partner will serve as the industrial deployment and validation lead, providing real-world platforms, operational expertise, and commercialization pathways. Key responsibilities include:
	• **Industrial Platform Provisioning**
	  ○ Provide access to industrial-grade robotic systems (e.g., AMRs, cobots)
	  ○ Supply factory-floor environments and representative manufacturing tasks
	  ○ Support integration with existing manufacturing infrastructure (e.g., MES, PLCs)
	• **System Integration and Deployment**
	  ○ Integrate Swarm Contract protocols with production-ready robotic control systems
	  ○ Adapt interfaces to meet industrial safety, reliability, and performance requirements
	  ○ Support deployment and operation of physical pilot systems
	• **Industrial Validation and Testing**
	  ○ Conduct pilot trials under real manufacturing conditions
	  ○ Evaluate system behavior under operational disturbances and workload variability
	  ○ Provide empirical feedback on performance, usability, and maintainability
	• **Engineering Feedback and Refinement**
	  ○ Identify practical constraints and requirements from factory operations
	  ○ Collaborate on protocol refinements to improve deployability and robustness
	  ○ Validate system performance against industry-relevant KPIs
	• **Commercialization and Scaling Pathways**
	  ○ Assess market applicability and customer needs
	  ○ Explore productization strategies and business models
	  ○ Support technology transfer and early-stage commercialization efforts
	  ○ Contribute to standardization discussions related to smart manufacturing and decentralized systems
	  ○ Support dissemination to industrial stakeholders and ecosystems

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected Technical Outcomes**
	• A validated Swarm Contract framework enabling decentralized coordination of heterogeneous robotic agents without centralized scheduling
	• A formally specified and auditable coordination protocol, including task allocation, adjudication, and settlement mechanisms
	• A scalable simulation environment for evaluating decentralized manufacturing workflows under realistic constraints
	• Demonstrated performance gains over centralized baselines, including:
	  ○ Reduced idle time
	  ○ Improved resource utilization
	  ○ Lower task completion latency
	• A physical multi-robot prototype implementing Swarm Contracts on real robotic platforms
	• Advanced coordination capabilities, including:
	  ○ Contract splitting and merging for complex tasks
	  ○ Dynamic adaptation to robot availability and task demand
	  ○ Quantitative evidence of robustness to heterogeneity, failures, and operational uncertainty
	
	Expected Industrial Outcomes
	• A deployable decentralized coordination technology suitable for smart manufacturing environments
	• Reduced reliance on costly, centralized manufacturing execution systems

- Lower integration barriers for factories with heterogeneous robots from multiple vendors
- Improved operational flexibility for high-mix, low-volume production scenarios
- Validated industrial pilot deployment demonstrating feasibility in real factory conditions
- Clear engineering guidelines and best practices for adoption by SMEs
- Foundations for commercial productization and technology transfer

Expected Economic and SME Impact

- Reduced upfront and maintenance costs for automation infrastructure
- Faster deployment timelines compared to centralized scheduling systems
- Increased accessibility of advanced automation technologies for SMEs
- Improved competitiveness of Korean manufacturing SMEs through:
  - Flexible production
  - Decentralized control
  - Resilient operations
- Long-term potential for application beyond manufacturing, including:
  - Logistics and warehousing
  - Disaster response
  - Distributed service robotics

Expected Scientific and Research Impact

- Advancement of the state of the art in:
  - Decentralized multi-agent systems
  - Swarm robotics
  - Smart contract–enabled coordination
- New empirical evidence bridging theory, simulation, and physical deployment
- Peer-reviewed publications, open research artifacts, and joint IP

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization